

EXAMPLE 10**Implied Return and Growth for a Stock**

Coca-Cola Company stock trades at a share price of USD63.00 and its annualized expected dividend per share during the next year is USD1.76.

1. If an analyst expects Coca-Cola's dividend per share to increase at a constant 4 percent per year indefinitely, calculate the required return expected by investors.

Solution:

6.79 percent

Using Equation 21,

$$r = \frac{1.76}{63} + 0.04 = 0.0679.$$

The required return expected for Coca-Cola stock is 6.79 percent given its current price, expected dividend, and expected dividend growth rate. Investor expectations of future stock returns are inferred by the combination of the current price, expected future cash flows and the cash flow growth rate. Suppose that, instead of attempting to estimate the required return, an investor wishes to determine an implied dividend growth rate. In this case, the investor must assume a future stock return, as in question 2.

2. If the analyst believes that Coca-Cola stock investors should expect a return of 7 percent, calculate the implied dividend growth rate for Coca-Cola.

Solution:

4.21 percent

Using Equation 22,

$$g = 0.07 - \frac{1.76}{63} = 0.0421.$$

The implied dividend growth rate for Coca-Cola stock is 4.21 percent given its expected return, price, and expected dividend. Given that a higher expected return is assumed in this question compared to the case in question 1, the result is a higher implied dividend growth rate to justify Coca-Cola's stock price of USD63.00.

Rather than comparing equity share prices directly in currency terms, a common practice is to compare ratios of share price to earnings per share, or the **price-to-earnings ratio**.

PRICE-TO-EARNINGS RATIO

Price-to-earnings ratio is a relative valuation metric that improves comparability by controlling for a known driver of value (earnings per share) as well as currency. It is analogous to expressing the price of real estate using a price per square meter.

A stock trading at a price-to-earnings ratio of 20 implies that its share price is 20 times its earnings per share and investors are willing to pay 20 times earnings per share for each share traded, which is more expensive than a stock trading at a price to earnings ratio of 10.